

Frank — AMC Medal Drill

25 Jun 2026 | Targeted on your 2018 gaps: cross-number / digit-property & organised counting | All answers are whole numbers. Show your working.

1. [5 marks] A box held 60 counters but some were lost. When the rest are shared equally among 4 people, 2 are left over; shared among 5 people, 4 are left over. Which of these could be the number lost: 4, 5, 6, 19, 23?

2. [6 marks] What is the smallest three-digit number that is even, a multiple of 6, contains the digit 2, has no digit 0, and has all different digits?

3. [6 marks] What is the smallest three-digit number that is even, a multiple of 6, contains the digit 3, has no digit 0, and has all different digits?

4. [6 marks] What is the smallest three-digit number that is even, a multiple of 4, contains the digit 5, has no digit 0, and has all different digits?

5. [7 marks] In how many ways can you choose 3 different numbers from 2 to 13 so that their sum is a multiple of 4?

6. [7 marks] In how many ways can you choose 3 different numbers from 4 to 15 so that their sum is a multiple of 2?

Every question is a fresh, code-verified twin of a real AMC Upper Primary question (no past question is reproduced). Tip: for “multiple of 9” add the digits; apply the strongest constraint first.